

Homework 2: k Sample Inference

Lions like to eat onions

1 Question 1

1.1 Part a

1.1.1 1

```
1 set.seed(354)
2
3 R <- 1000
4 simulations <- vector("list", R)
5
6 for (i in 1:R) {
7   simulations[[i]] <- data.frame(
8     value = rnorm(150, mean = 0, sd = 1),
9     group = factor(rep(1:5, each = 30))
10  )
11 }
```

1.1.2 2

```
1 error_unadjusted <- logical(R)
2
3 for (i in 1:R) {
4   dat <- simulations[[i]]
5
6   test_unadjusted <- pairwise.t.test(
7     dat$value, dat$group,
8     p.adjust.method = "none"
9   )
10
11   error_unadjusted[i] <- any(
12     test_unadjusted$p.value < 0.05,
13     na.rm = TRUE
14   )
15 }
16
17 mean(error_unadjusted)
```

```
[1] 0.294
```

At least one pairwise comparison provided statistically discernible support for a non-zero mean difference in 294 of the 1,000 simulations ($p < 0.05$). These are Type I errors, giving an estimated family-wise error rate of 29.4%.

1.1.3 3

```
1 error_tukey <- logical(R)
2
```

```

3 for (i in 1:R) {
4   dat <- simulations[[i]]
5
6   fit <- aov(value ~ group, data = dat)
7   test_tukey <- rstatix::tukey_hsd(fit)
8
9   error_tukey[i] <- any(test_tukey$p.adj < 0.05)
10 }
11
12 mean(error_tukey)

```

```
[1] 0.053
```

After Tukey's HSD adjustment, at least one pairwise comparison provided statistically discernible support for a non-zero mean difference in 53 of the 1,000 simulations (adjusted $p < 0.05$). Since all population means were zero, these are Type I errors, and the estimated family-wise error rate is 5.3%.

1.2 Part b

```

1 error_bh <- logical(R)
2
3 for (i in 1:R) {
4   dat <- simulations[[i]]
5
6   test_bh <- pairwise.t.test(
7     dat$value, dat$group,
8     p.adjust.method = "BH"
9   )
10
11   error_bh[i] <- any(
12     test_bh$p.value < 0.05,
13     na.rm = TRUE
14   )
15 }
16
17 mean(error_bh)

```

```
[1] 0.047
```

After BH adjustment, at least one pairwise comparison provided statistically discernible support for a non-zero mean difference in 47 of the 1,000 simulations, giving an estimated family-wise error rate of 4.7%. This was much lower than the unadjusted rate of 29.4% and close to Tukey's rate of 5.3%.

BH targets the false discovery rate (FDR), whereas Tukey's HSD controls the family-wise error rate (FWER).

2 Question 2

2.1 Part a

```

1 # Required sample size per group
2 pwr::pwr.anova.test(
3   k = 5,
4   f = 0.25,
5   sig.level = 0.05,
6   power = 0.80
7 )

```

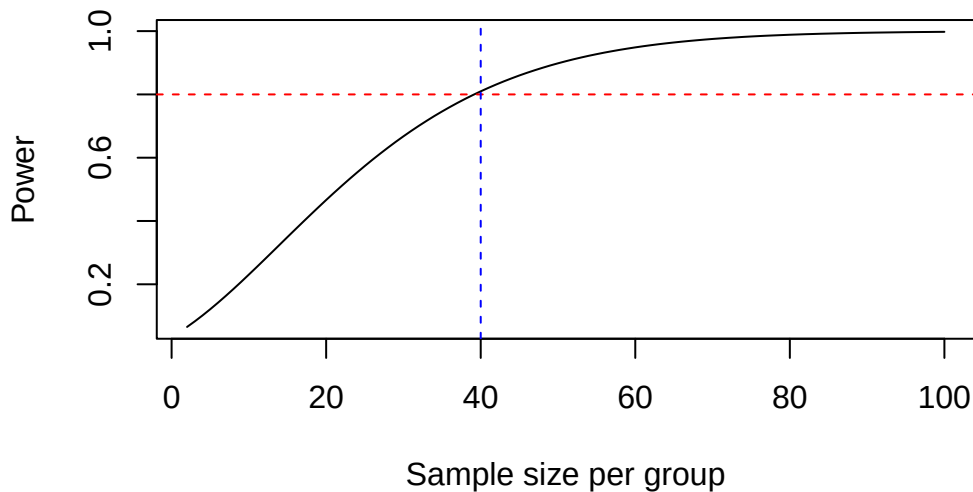
Balanced one-way analysis of variance power calculation

```
k = 5
n = 39.1534
f = 0.25
sig.level = 0.05
power = 0.8
```

NOTE: n is number in each group

```
1 # Power curve
2 n <- 2:100
3
4 power <- sapply(n, function(x) {
5   pwr::pwr.anova.test(
6     k = 5, n = x, f = 0.25, sig.level = 0.05
7   )$power
8 })
9
10 plot(n, power, type = "l",
11       xlab = "Sample size per group",
12       ylab = "Power",
13       main = "ANOVA Power Curve")
14
15 abline(h = 0.80, col = "red", lty = 2)
16 abline(v = 40, col = "blue", lty = 2)
```

ANOVA Power Curve



The required sample size is roughly 40 participants per group and 200 in total. The graph shows that 40 participants per group achieves slightly more than 80% power.

Since every group in the original groups had more than 40 participants, the study is powered for the overall ANOVA under the assumed moderate effect size. This calculation assumes equal group sizes, so it does not provide the exact power of the original unequal-sized design.

2.2 Part b

```
1 # Non-centrality parameter and power
2 lambda <- 150 * 0.25^2
3 lambda
```

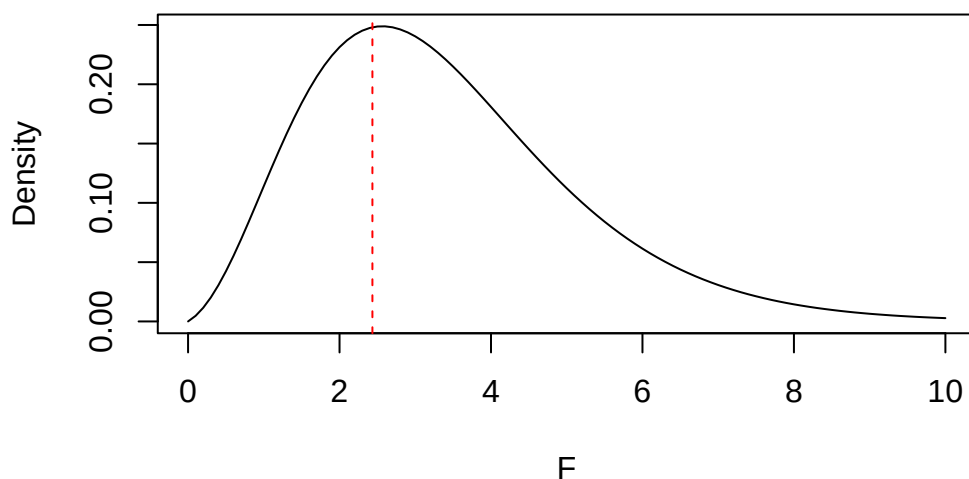
```
[1] 9.375
```

```
1 critical <- qf(0.95, 4, 145)
2 pf(critical, 4, 145, ncp = lambda, lower.tail = FALSE)
```

```
[1] 0.6675964
```

```
1 # Alternative F distribution
2 curve(df(x, 4, 145, ncp = lambda), from = 0, to = 10,
3       xlab = "F", ylab = "Density",
4       main = "Alternative F Distribution: N = 150")
5 abline(v = critical, col = "red", lty = 2)
```

Alternative F Distribution: N = 150



With 30 participants per group, the non-centrality parameter is 9.375 and the power is approximately 66.8%. Therefore, $N = 150$ is insufficient to achieve the desired 80% power.

2.3 Part c

```
1 # Power for different sample sizes per group
2 n <- 40:500
3
4 power_c <- sapply(n, function(x) {
5   PMCMRplus::power.tukey.test(
6     n = x,
7     groups = 5,
8     delta = 0.25,
9     within.var = 0.7152,
10    sig.level = 0.05
11   )$power
12 })
13
14 # Power with 40 participants per group
15 power_c[1]
```

```
[1] 0.07866837
```

```
1 # Smallest sample size achieving at least 80% power
2 n_required <- min(n[power_c >= 0.80])
3 n_required
```

```
[1] 293
```

An overall ANOVA can detect differences among groups having different means. However, detecting the smaller difference between stages 1 and 2 is harder, especially after Tukey's adjustment for multiple comparisons.

With 40 participants per group, the power is only about 7.9%. Using the pooled variance of 0.7152, approximately 293 participants per group are needed to achieve 80% power for this comparison.

2.4 Part d

```

1 library(Superpower)
2
3 for (n in c(40, 80)) {
4
5   design <- ANOVA_design(
6     design = "5b",
7     n = n,
8     mu = c(2.92, 3.11, 3.36, 4.04, 4.35),
9     sd = sqrt(0.80),
10    plot = FALSE
11  )
12
13  result <- ANOVA_power(
14    design,
15    nsims = 1000,
16    seed = 354,
17    emm = TRUE,
18    emm_p_adjust = "tukey",
19    verbose = FALSE
20  )
21
22  print(paste("Sample size per group:", n))
23  print(result$main_result)
24  print(result$emm_results)
25 }

```

```

[1] "Sample size per group: 40"
      power effect_size
anova_a  100  0.2875741
      contrast power    cohen_f
p_1  a_a1 - a_a2   3.2 0.07851998
p_2  a_a1 - a_a3  29.8 0.16043752
p_3  a_a1 - a_a4  99.7 0.40079656
p_4  a_a1 - a_a5 100.0 0.51421433
p_5  a_a2 - a_a3   6.4 0.09965935
p_6  a_a2 - a_a4  96.8 0.33441403
p_7  a_a2 - a_a5  99.9 0.44783180
p_8  a_a3 - a_a4  73.3 0.24120424
p_9  a_a3 - a_a5  98.1 0.35462200
p_10 a_a4 - a_a5  11.9 0.11742044
[1] "Sample size per group: 80"
      power effect_size
anova_a  100  0.2797892
      contrast power    cohen_f
p_1  a_a1 - a_a2   8.2 0.07492609
p_2  a_a1 - a_a3  67.3 0.15861698
p_3  a_a1 - a_a4 100.0 0.39912272
p_4  a_a1 - a_a5 100.0 0.51008441
p_5  a_a2 - a_a3  18.1 0.09050315

```

```
p_6  a_a2 - a_a4 100.0 0.32897230
p_7  a_a2 - a_a5 100.0 0.43993399
p_8  a_a3 - a_a4  98.4 0.24067913
p_9  a_a3 - a_a5 100.0 0.35164081
p_10 a_a4 - a_a5  30.1 0.11153257
```

Doubling the sample size from 40 to 80 per group increases power, but is still insufficient to achieve 80% power for every pairwise comparison. In specific, the small difference between stages 0 and 1 ($3.11 - 2.92 = 0.19$) remains difficult to detect.

3 Question 3

3.1 Part a

Let μ_{2C} , μ_{6C} , μ_{16C} , and μ_{16T} denote the population mean SUVmax values for the 2-month control, 6-month control, 16-month control, and 16-month treatment groups, respectively.

$$H_0 : \mu_{2C} = \mu_{6C} = \mu_{16C} = \mu_{16T}$$

$$H_a : \text{At least one population mean differs.}$$

3.2 Part b

```
1 dat3 <- data.frame(
2   SUVmax = c(
3     1.48969, 1.30168, 1.30620,
4     4.36000, 4.28807, 3.64184,
5     7.30182, 7.12416, 8.28764,
6     4.69304, 2.51326, 2.38019
7   ),
8   group = factor(
9     rep(c("2M Control", "6M Control",
10          "16M Control", "16M Treatment"), each = 3),
11     levels = c("2M Control", "6M Control",
12                "16M Control", "16M Treatment")
13   )
14 )
15
16 # Means and standard deviations
17 aggregate(SUVmax ~ group, data = dat3, FUN = mean)
```

```
      group  SUVmax
1  2M Control 1.365857
2  6M Control 4.096637
3 16M Control 7.571207
4 16M Treatment 3.195497
```

```
1 aggregate(SUVmax ~ group, data = dat3, FUN = sd)
```

```
      group  SUVmax
1  2M Control 0.1072666
2  6M Control 0.3955041
3 16M Control 0.6267761
4 16M Treatment 1.2986162
```

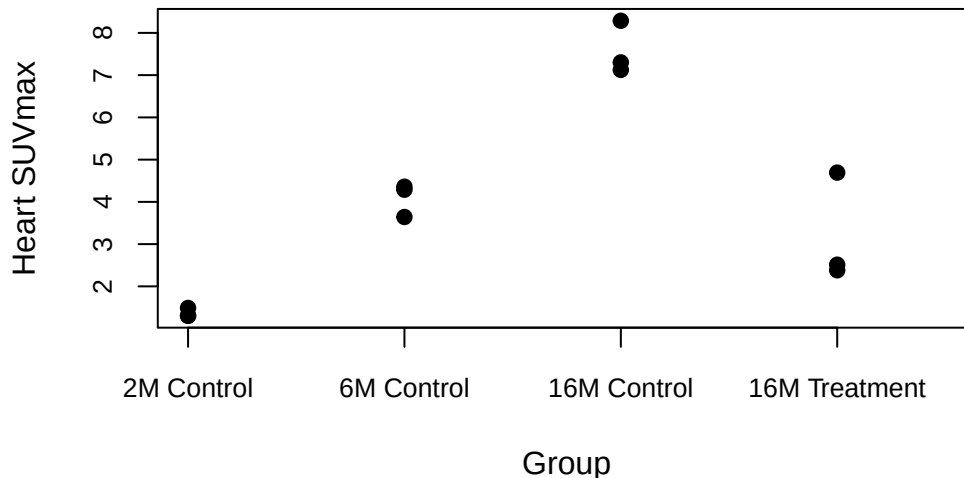
```
1 # Display all observations
2 stripchart(SUVmax ~ group, data = dat3,
```

```

3     vertical = TRUE, method = "stack", pch = 19,
4     xlab = "Group", ylab = "Heart SUVmax",
5     main = "Heart SUVmax by Age and Treatment",
6     cex.axis = 0.8)

```

Heart SUVmax by Age and Treatment



Among the control groups, mean heart SUVmax increased with age (approximately 1.37, 4.10, and 7.57 for 2, 6, and 16 months, respectively). The 16-month treatment group had a lower mean (approximately 3.20) than the 16-month control group, placing it between the two younger control groups.

3.3 Part c

```

1 test3 <- oneway.test(SUVmax ~ group, data = dat3)
2 test3

```

One-way analysis of means (not assuming equal variances)

data: SUVmax and group

F = 98.464, num df = 3.0000, denom df = 3.5496, p-value = 0.0006851

SUVmax is quantitative. The sample standard deviations differ substantially across groups, so I used Welch's ANOVA because it does not assume equal population variances. The rats are not connected in certain ways so I assume independence. The observations per group are three, which is low, so the results should be interpreted cautiously.

Welch's ANOVA gives $F = 98.46$, with 3 and 3.55 degrees of freedom ($p = 6.85 \times 10^{-4}$). Since $p < 0.05$, we reject the null hypothesis. The data provide statistically discernible support that at least one population mean heart SUVmax differs among the four groups.

3.4 Part d

```

1 posthoc3 <- rstatix::games_howell_test(
2   dat3, SUVmax ~ group
3 )
4
5 posthoc3

```

A tibble: 6 x 8

.y.	group1	group2	estimate	conf.low	conf.high	p.adj	p.adj.signif
* <chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>

1	SUVmax	2M	Control	6M	Control	2.73	1.31	4.15	0.0119	*
2	SUVmax	2M	Control	16M	Contr~	6.21	3.81	8.60	0.00720	**
3	SUVmax	2M	Control	16M	Treat~	1.83	-3.30	6.96	0.308	ns
4	SUVmax	6M	Control	16M	Contr~	3.47	1.56	5.39	0.00801	**
5	SUVmax	6M	Control	16M	Treat~	-0.901	-5.47	3.67	0.698	ns
6	SUVmax	16M	Control	16M	Treat~	-4.38	-8.51	-0.244	0.0431	*

Games–Howell comparisons showed statistically discernible increases in mean heart SUVmax across the control groups: 2 versus 6 months (adjusted $p = 0.0119$), 2 versus 16 months ($p = 0.0072$), and 6 versus 16 months ($p = 0.00801$). The 16-month treatment group had a lower mean than the 16-month control group (estimated difference = -4.38 , adjusted $p = 0.0431$), supporting a reduction associated with low-dose nicotine.

The treatment group did not differ significantly from the 2-month or 6-month control groups (adjusted $p = 0.308$ and 0.698 , respectively), but these results do not establish equivalence to younger mice.

3.5 Part e

The results support the hypothesis that SUVmax increases with age and is reduced by low-dose nicotine in older mice. All three control groups differed significantly, with higher mean SUVmax at older ages. Although the treatment group did not differ significantly from the younger controls, this does not establish that treatment restored SUVmax to younger levels.